

Channeling Density: Sewerage Investments in 19th-Century Paris

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Channeling density

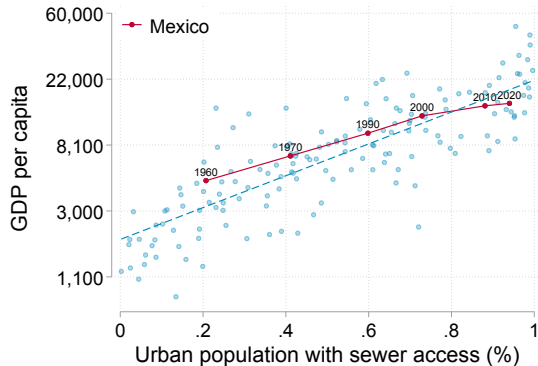
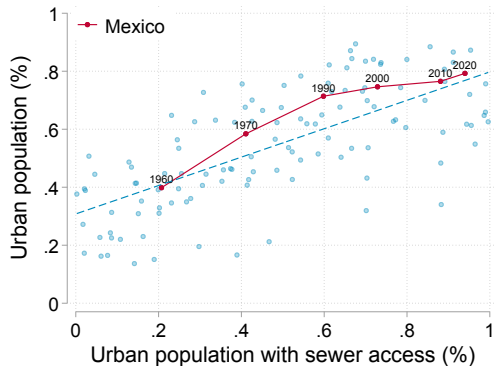
- Cities provide services and promise **agglomeration externalities**
- But urban density gives rise to negative sanitation externalities
- These **congestion externalities** may constrain more efficient allocations



- Sanitation investments may reduce congestion and facilitate agglomeration

How do sewerage investments affect congestion externalities? GE?

Sewer access correlates with development



Note: country-year observations from IPUMS

What we do to answer our question

How do sewerage investments affect congestion externalities? GE?

- **Setting:** 19th century Paris
- **Data:** newly digitalised address level data
- **Empirical strategy:** event studies & engineering IV
- **Reduced form:** sewers $\Rightarrow$ population, firm counts, house prices, typhoid deaths
- **Quantitative analysis:**
 1. Estimate parcel level amenity and productivity
 2. Regress them on: sewers, population, sewers $\times$ population
- **Counterfactuals:** examine impact of sewers & assess importance of externality

Contributions to the literature

Empirical studies on sanitation and health + urban outcomes

Cutler and Miller (2005), Anderson et al (2022), Costa and Kahn (2015), Kestzenbaum and Rosenthal (2017), Alsan and Goldin (2019), Ambrus et al (2020), Heblich et al (2021), Coury et al. (2024), McCulloch et al (2025)

- ▶ Quantitative urban model
- ▶ Evidence on employment response & long-run adjustments

Quantitative studies on (historical) development of cities

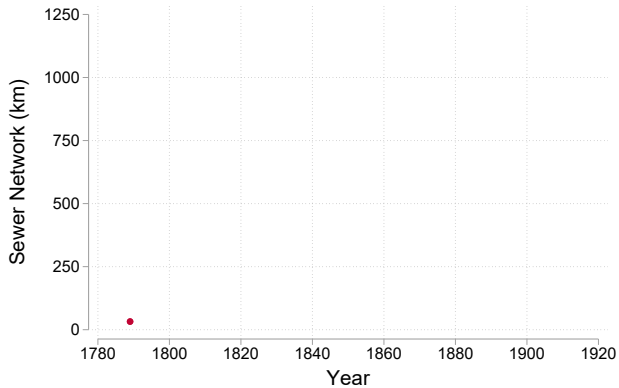
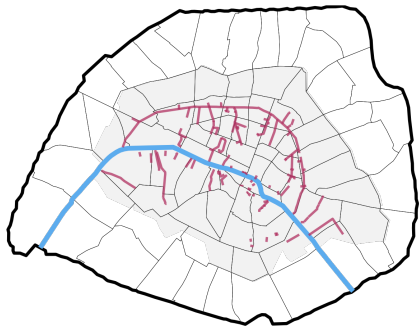
Ahlfeldt et al (2015), Heblich et al (2020), Redding and Sturm (2024)

- ▶ Examine importance of sanitation congestion externality

Background

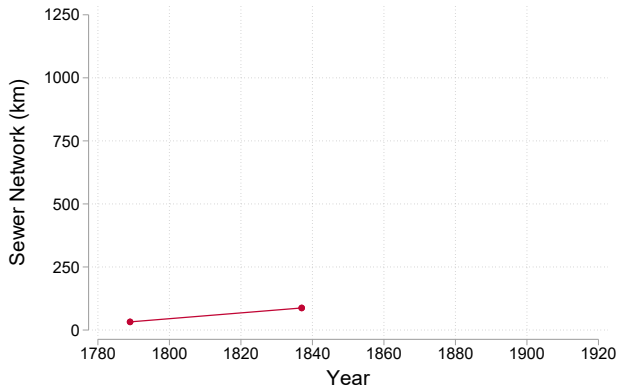
Sewer network expansion

1789



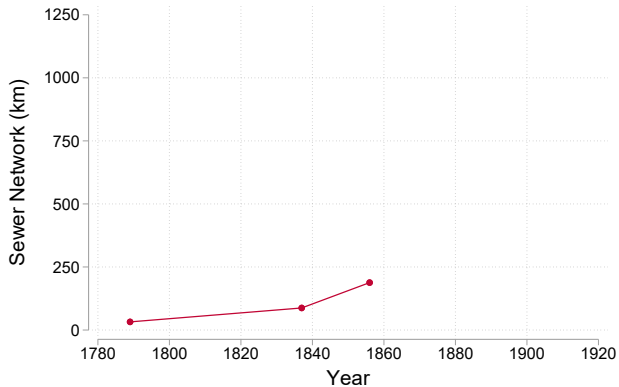
Sewer network expansion

1837



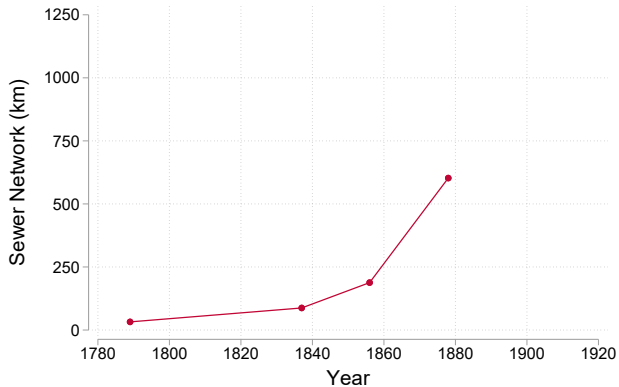
Sewer network expansion

1856



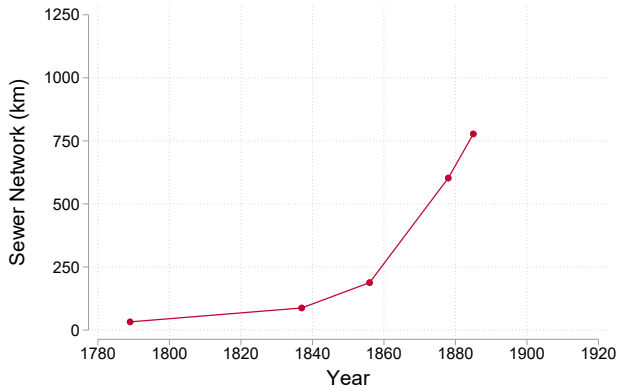
Sewer network expansion

1878



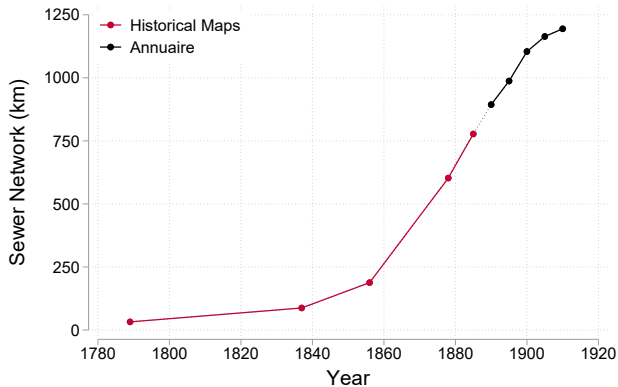
Sewer network expansion

1885



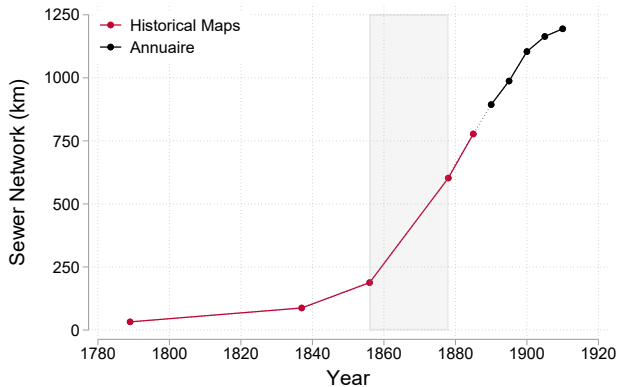
Sewer network expansion

1885



Sewer network expansion

1885

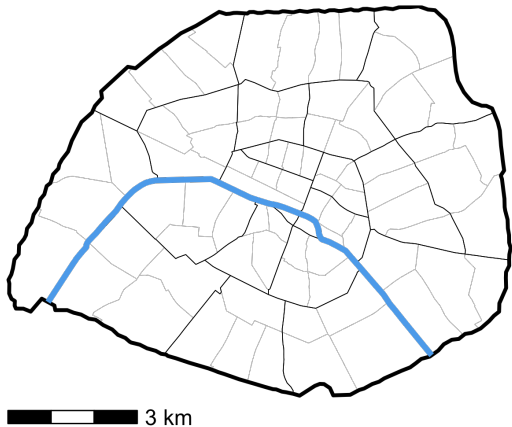


The Parisian sewers

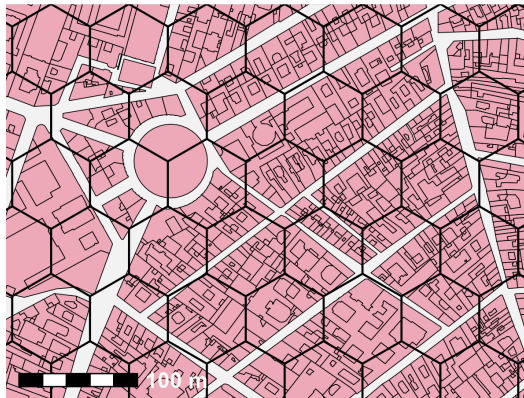
- Poor sanitary conditions in early 19th-century Paris
 - ▶ Wastewater flowed through street and remaining stagnant
 - ▶ Open defecation and polluted wells
- Sewers = combined system for stormwater and wastewater
- Direct building connections only permitted after 1886
- Sewer network expansion was part of Haussmann's renovation
 - ▶ Other constructions: parks & boulevards

Geography & Data

Geography



⇒ 20 arrondissements & 80 quartiers



⇒ over 10,000 hexagons

Geolocated historical database

We digitised a **new historical address-level dataset** spanning a 100 years on:

1. Infrastructure
2. Population
3. Firms
4. House prices
5. Typhoid deaths

Reduced Form Evidence

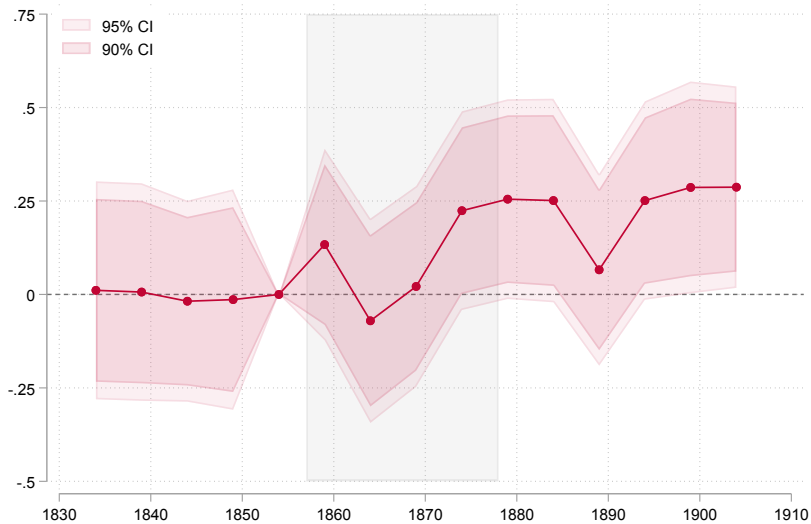
House prices: event study specification

$$\ln(\text{Price}_{ht}) = \delta_h + \delta_t + \sum_t \beta_t T_h \times D_t + \delta_{qd} + \theta \text{MainRoad}_h \times \text{Post}_t + \varepsilon_{ht}$$

- Where
 - ▶ h : hexagon
 - ▶ t : 5-year period
 - ▶ δ_{qd} : quartier-decade FE
- Treatment status (T_h) defined by corresponding hexagon
 - ▶ Drop hexagons with sewers in 1856
 - ▶ **Treated**: 1857-1878 cohort
 - ▶ **Control**: no sewer in 1878



Log house prices



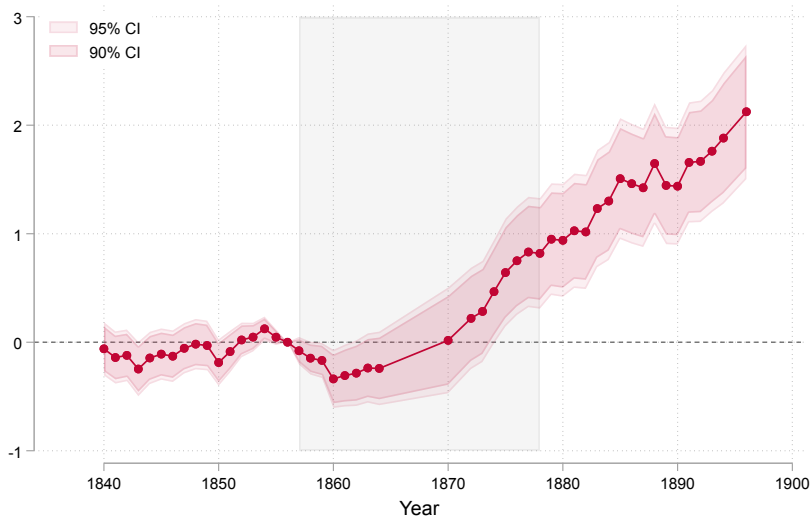
Firms: event study specification

$$\text{FirmCount}_{ht} = \delta_h + \delta_t + \sum_t \beta_t T_h \times D_t + \delta_{qd} + \theta \text{MainRoad}_h \times \text{Post}_t + \varepsilon_{ht}$$

- Where
 - ▶ h : hexagon
 - ▶ t : year
 - ▶ δ_{qd} : quartier-decade FE
- Treatment status (T_h) defined by corresponding hexagon
 - ▶ Drop hexagons with sewers in 1856
 - ▶ **Treated**: 1857-1878 cohort
 - ▶ **Control**: no sewer in 1878



Firm count

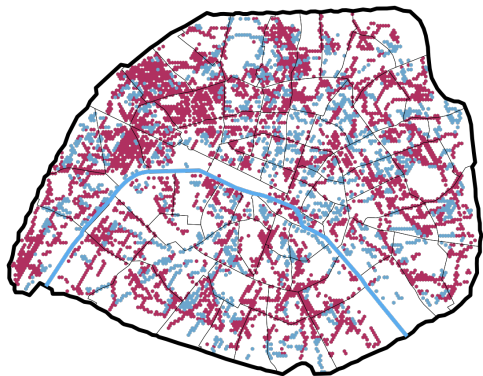


Note: control mean in 1878 is 5.83

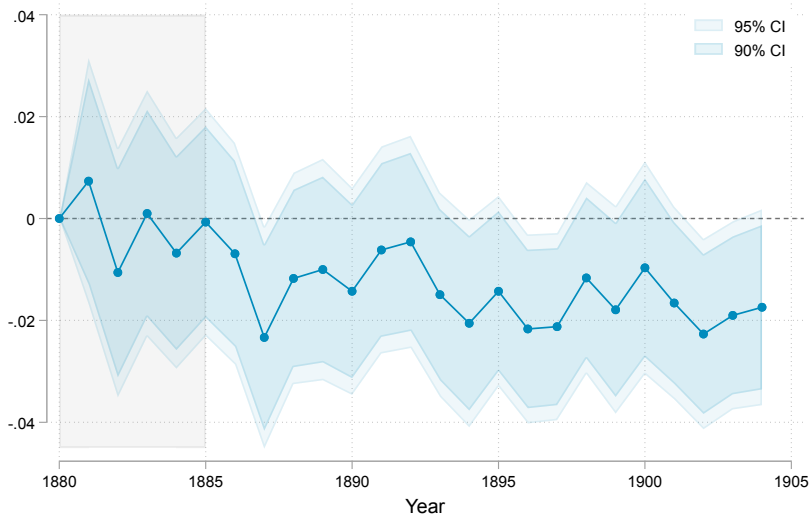
Typhoid deaths: event study specification

$$\mathbb{1}\{\text{Typhoid death}\}_{ht} = \delta_h + \delta_t + \sum_t \beta_t T_h \times D_t + \delta_{qd} + \varepsilon_{ht}$$

- Where
 - ▶ h : hexagon
 - ▶ t : year
 - ▶ δ_{qd} : quartier-decade FE
- Treatment status (T_h) defined by corresponding hexagon
 - ▶ Drop hexagons with sewers in 1856 and hexagons without sewers in 1885
 - ▶ **Treated**: 1879-1885 cohort
 - ▶ **Control**: 1857-1878 cohort



Probability typhoid death



Note: control mean in 1880 is 0.09

Quantitative Analysis

Including sewers into the canonical urban model

- Indirect utility of o living in location i and working in location j is

$$u_{ij o} = \frac{B_i w_j z_{ij o}}{d_{ij} (P_i^h)^{1-\beta}}$$

- ▶ Amenities B_i , wage w_j , idiosyncratic component $z_{ij o}$
 - ▶ Commuting costs d_{ij} , residential housing P_i^h
- Amenities B_i depend on sewer access and population

$$\ln(B_i) = \delta_a + \eta_1 \text{Sewer}_i + \eta_2 \ln(R_i) + \eta_3 \text{Sewer}_i \times \ln(R_i) + X_i \theta + b_i$$

- ▶ Residential population R_i , sewer dummy Sewer_i , neighborhood FEs δ_a , controls X_i
- Analogous expression for location specific productivity A_i

Quantitative analysis

Goal is to estimate sewerage related parameters η

Step 1: solving for wages

Step 2: backing out amenity and productivity

Step 3: estimating sewerage parameters

Provisional data inputs

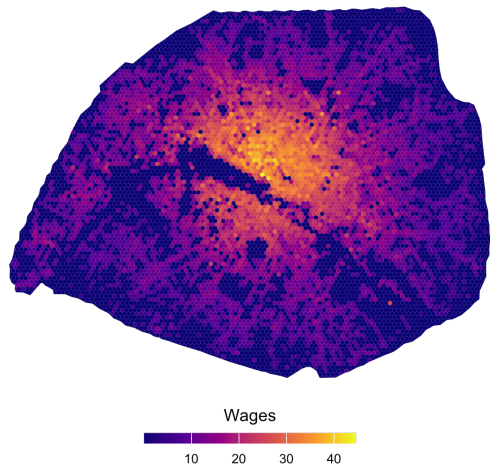
- Population: quartier population distributed uniformly over build-up area
- Commuting times: straight line distances

Step 1: solving for wages

- Commuter market clearing conditions

$$L_j = \sum_s \frac{\left(\frac{w_j}{\tau_{sj}^{\kappa}}\right)^\epsilon}{\sum_k w_k \left(\frac{w_k}{\tau_{sk}^{\kappa}}\right)^\epsilon} R_s$$

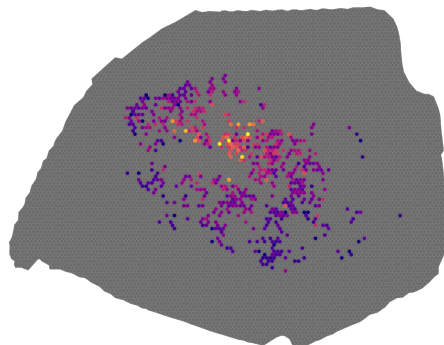
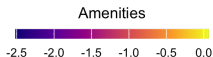
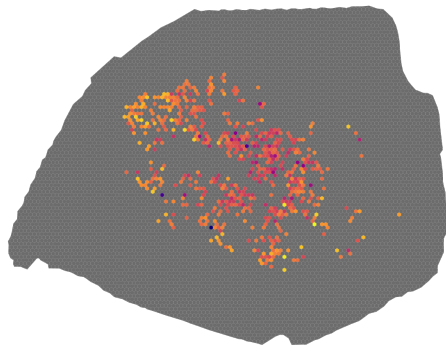
- ▶ Employment population L_i
- ▶ Residential population R_i
- ▶ Travel times τ_{ij}



Step 2: backing out amenities and productivity

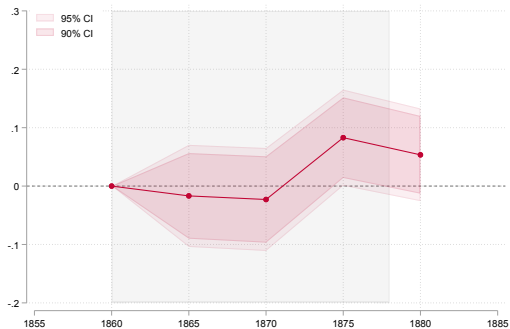
$$\ln(B_i) - \frac{1}{\epsilon} \ln(\Phi) = \frac{1}{\epsilon} \ln(\pi_i) + (1 - \beta) \ln(P_i^h) - \frac{1}{\epsilon} \ln \left[\sum_k \left(\frac{w_k}{\tau_{ik}^\kappa} \right)^\epsilon \right]$$

$$\ln(A_j) = \alpha \ln \left(\frac{w_j}{\alpha} \right) + (1 - \alpha) \ln \left(\frac{q_j}{1 - \alpha} \right)$$

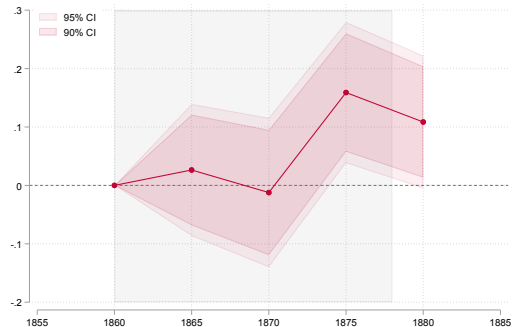


Step 3: mimicking the event studies

Amenities



Productivity



Step 3: estimating sewerage parameters

- Estimating equation for amenities

$$\Delta \ln(B_i) = \tilde{\delta}_a + \eta_1 \Delta \text{Sewer}_i + \eta_2 \Delta \ln(R_i) + \eta_3 \Delta \text{Sewer}_i \times \Delta \ln(R_i) + X_i \theta + \tilde{b}_i$$

- Analogous expression for productivity
- Instruments
 - ▶ *Sewer access*: based on engineering costs of connecting a location
 - ▶ *Population*: based on transport network expansion
 - ▶ Interaction of the above

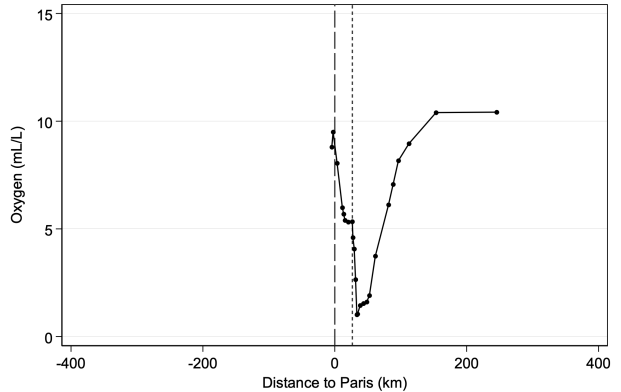
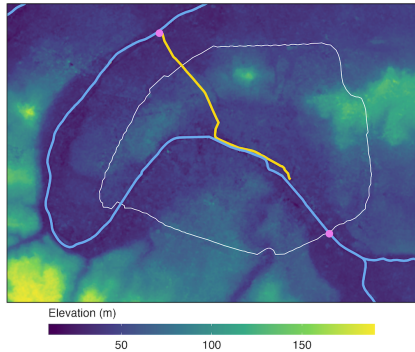
Next Steps

Next steps

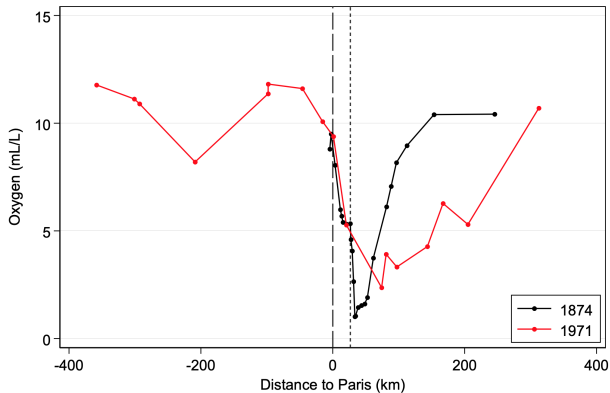
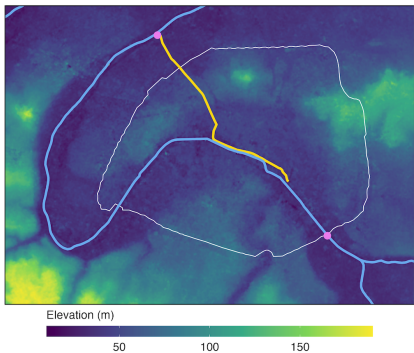
- **Data:** keep digitalising population and house price records
- **Reduced form:** examine spillovers and heterogeneity
- **Quantitative analysis:** prepare instruments

What happened to the sewage?

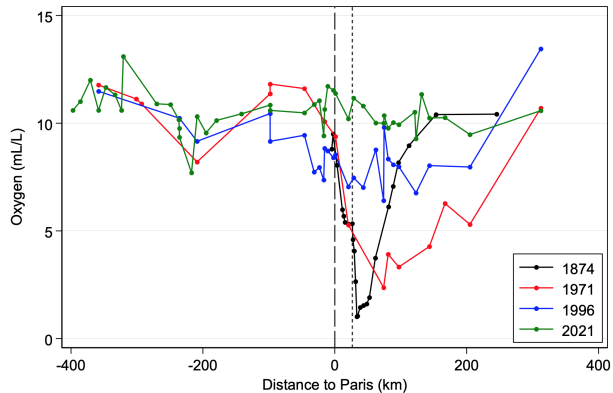
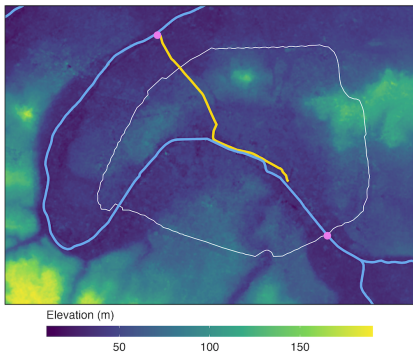
Sewage pollution of the Seine



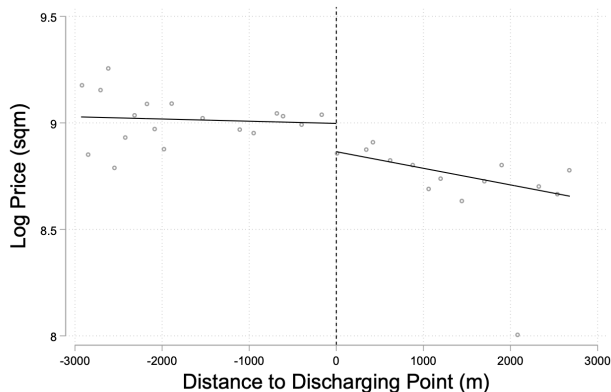
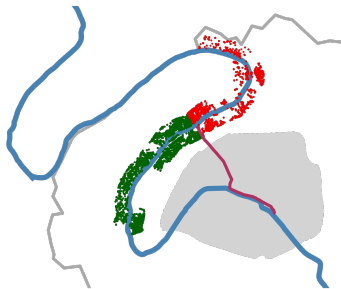
Sewage pollution of the Seine



Sewage pollution of the Seine



RDD: house prices along the Seine



⇒ using optimal bandwidth: 13% house price drop today significant at the 1% level